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Live Summarization of Surveillance Streams: A Retrieval Augmented Vision-Language Agent using YOLOv8, LLaVA, and LangChain

Group No. 11

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Project Guide

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Abstract

- Traditional retail surveillance systems are reactive and depend on manual review of hours-long video footage.
- This process is inefficient, time-consuming, and often fails to provide actionable insights in real time.
- Our project proposes an end-to-end intelligent system that can perform live summarization and querying of surveillance streams.
- The system integrates YOLOv8 for real-time detection, Deep SORT for tracking, and LLaVA for generating textual summaries of events.
- Summaries are stored as vector embeddings in ChromaDB, enabling semantic search with metadata like timestamps.
- A LangChain-based conversational AI allows managers to query incidents in natural language and get instant, evidence-backed answers.
- This transforms passive surveillance into an interactive, intelligent tool that improves retail security and operational efficiency.

Sustainable Development Goals(SDG) mapped

SDG 11: Sustainable Cities and Communities
Intelligent surveillance enhances safety and security in commercial spaces.

Introduction

- Traditional retail surveillance relies on manual review of video, making it time-consuming and reactive.
- There is a need for systems that can provide real-time, actionable insights from live video feeds.
- Our project introduces an AI-powered surveillance summarization system that integrates computer vision, language models, and vector databases.
- This transforms surveillance into an intelligent, interactive tool for improving security and efficiency.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
1	Smart CCTV Video Summarization: A Deep Learning Fusion of YOLO and DeepSORT Algorithms	Vaishali Jabade, Aditya Ingale, Rushikesh Joshi, Gajanan Kulkarni	2024	Proposed a CCTV summarization framework using YOLOv5 for real-time object detection and Deep SORT with Kalman filtering for object tracking. Frames with dynamic elements are merged to produce concise video summaries.	Limited to YOLOv5 and Deep SORT; lacks evaluation on large-scale, diverse real-world datasets.
2	Detection of Criminal Activities Through CCTV by Live Footage Analysis	Rahul Kumar, Shilpi Sharma, Ayan Sar, Tanupriya Choudhury	2024	Developed a YOLO-based CCTV crime detection system for theft,vandalism, and violent behavior. Integrated deep learning with real-time alerting, transfer learning, and data augmentation for adaptability	High computational demand; privacy and ethical concerns; performance may drop in poor lighting or crowded scenes.
3	Video Summarization Techniques: A Comprehensive Review	Toqa Alaa, Ahmad Mongy, Assem Bakr, Mariam Diab, Walid Gomaa	2024	Survey of extractive and abstractive summarization methods, including clustering, deep learning (CNN, RNN, LSTM, Transformers), attention, GANs, reinforcement learning, and multi-modal approaches.	Being a review, no implementation; highlights research gaps such as dataset limitations and real-world deployment issues.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
4	Recent Challenges and Opportunities in Video Summarization with Machine Learning Algorithms	Payal Kadam, Shruti Patil, Deepali Vora, Sashikala Mishra, Ketan Kotecha, Ajith Abraham, Lubna Abdelkareim Gabralla	2022	Comprehensive survey of video summarization methods from 2005–2022. Categorized techniques into static vs dynamic, supervised vs unsupervised, single-view vs multi-view. Discussed datasets (TVSum, SumMe) and metrics (F-score, precision, recall). Highlighted need for robust MVS approaches and advanced deep architectures like3D-CNNs and Conv-LSTMs.	Survey only, no implementation; MVS methods remain underexplored and face synchronization and real-world challenges.
5	Automatic Video Summarization with Timestamps using Natural Language Processing Text Fusion	Ahmed Emad, Fady Bassel, Mark Refaat, Mohamed Abdelhamed, Nada Shorim, Ashraf AbdelRaouf	2021	Proposed a multimodal system combining three modalities: visual analysis (frames), facial emotion recognition (CNN), and audio-to-text transcription. Outputs fused via NLP to generate a coherent textual summary with timestamps. Human evaluation achieved ~87% agreement on summary accuracy.	Dependent on quality of NLP fusion; struggles with silent or emotionless scenes; requires further refinement for natural sentence flow.
6	Intelligent Video Surveillance Based on YOLO: A Comparative Study	Rakesh Garg, Someet Singh	2021	Compared YOLOv3, YOLOv4, and YOLOv5 for handheld weapon detection in surveillance. YOLOv4 achieved the highest mAP (98.79%) and recall. Also tested tiny versions for real-time deployment.	YOLOv3/v4 are computationally heavy; tiny models reduce accuracy; lacks testing in varied environmental conditions.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
7	Unsupervised Video Summarization with a Convolutional Attentive Adversarial Network	Guoqiang Liang, Yanbing Lv, Shucheng Li, Shizhou Zhang, Yanning Zhang	2021	Uses a generator with fully convolutional sequence and attention mechanism; discriminator distinguishes summary-weighted features via GAN loss	Additional complexity from attention layers; potential overfitting on small datasets.
8	YOLO-Based Video Processing for CCTV Surveillance	Sujata Terdal, Sayyada F., Amulya Reddy, Manasi Koppal, Ameena Fatima	2020	Developed real-time CCTV surveillance using YOLOv3 for hazardous/suspicious object detection. Preprocessing included denoising, grayscale conversion, and resizing. Achieved 98.3% detection accuracy with high precision and recall (1.00 for "Person").	Focuses only on object detection, not activity recognition; scalability in crowded or multi-camera setups not addressed.
9	YOLO Based Real-Time Human Detection for Smart Video Surveillance at the Edge	Huy Hoang Nguyen, Van Truong Bui, Thi Nhung Ta, Hung Manh Pham	2020	Proposed a lightweight YOLOv2-based human detector optimized with residual blocks and SPP for edge devices. Achieved ~95% accuracy on INRIA and PENN-Fudan datasets; 2 FPS on Raspberry Pi 3B.	Limited processing speed (2 FPS); trade-off between accuracy and speed may restrict real-time use in high-density scenarios.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
10	Comprehensive Video Understanding: Video Summarization with Content-Based Video Recommender Design	Yudong Jiang, Kaixu Cui, Bo Peng, Changliang Xu	2019	Proposed a deep neural network (HighlightNet) treating video summarization as a content-based recommender problem. Integrated raw, semantic, audio, and vision features with GRU modeling, self-supervised learning, and data augmentation.	Requires large labeled datasets; subjectivity of summaries not fully addressed; risk of overfitting with limited data.
11	Cycle-SUM: Cycle-Consistent Adversarial LSTM Networks for Unsupervised Video Summarization	Li Yuan, Francis E. H. Tay, Ping Li, Li Zhou, Jiashi Feng	2019	Bi-LSTM selector + cycle-consistent dual GAN evaluator to preserve mutual information between summary and original video	More complex architecture; may require extensive training and tuning.
12	Deep Learning Approach for Suspicious Activity Detection from Surveillance Video	C. Rami Reddy, A. Sreenivasulu	2018	Implemented CNNs for feature extraction and LSTMs for temporal analysis to detect suspicious activities in CCTV. Used VGG-16 pretrained on ImageNet for behavior inference.	Tested on limited campus datasets; real-world scalability and accuracy in complex environments not validated.

Literature Review

Sr.no	Title	Author(s)	Year	Methodology	Drawback
13	Unsupervised Video Summarization with Adversarial LSTM Networks	Behrooz Mahasseni, Michael Lam, Sinisa Todorovic	2017	GAN framework with LSTM autoencoder summarizer and LSTM discriminator for unsupervised keyframe selection	May generate visually faithful but semantically weak summaries, struggles with long/complex videos and subjective variations.
14	Rapid Video Summarization on Compressed Video	Jurandy Almeida, Ricardo da S. Torres, Neucimar J. Leite	2010	Utilizes compressed-domain indicators such as I-frame DC images and HSV histograms to generate summaries without fully decoding video.	Performance degrades in scenes with dynamic or complex backgrounds; limited by reliance on compressed-domain features.
15	A Video Summarization Approach Based on Machine Learning	Wei Ren, Yuesheng Zhu	2008	Developed a machine learning model for automatic video transition prediction using features like cut, fade, dissolve, and pixel likelihood ratio. Keyframes are selected via a coarse-to-fine strategy to improve summarization.	Limited feature set; performance depends on shot transitions, less effective for long panning or tilting shots.

Research Gap(Limitations of existing systems)

- Limited testing on small or synthetic datasets, lacking real-world retail validation.
- High computational demand restricts real-time and edge deployment.
- Accuracy drops in low-light, crowded, or multi-camera environments.
- Focus mainly on object detection, not on activities or context.
- Generated summaries are often incomplete or incoherent.
- No support for intelligent natural language querying of incidents.
- Scalability issues for large-scale, real-time retail use.

Motivation

- Retail industries face major challenges like theft, fraud, and inefficiencies in operations.
- Managers often lack tools to quickly extract key information from large amounts of video data.
- Advances in vision-language AI models provide an opportunity to automate and enhance surveillance analysis.
- Our motivation is to bridge the gap between raw video streams and meaningful insights for decision-making.

Objectives

1. Develop a real-time system for detecting and tracking retail staff and shoppers by implementing a computer vision pipeline.

2. Create automated, context-aware summaries of key retail events using a Vision-Language Model to translate visual information into text.

3. Construct a searchable semantic database for loss prevention audits by storing vectorized event summaries and metadata.

4. Enable natural language queries of surveillance data using a Retrieval Augmented Generation (RAG) framework.

Problem Definition

- Current retail surveillance systems are reactive, requiring hours of manual review of video footage to identify incidents.
- This results in delays, inefficiency, and missed critical events, limiting their usefulness in preventing theft or improving operations.
- Managers do not have an easy way to search or query past incidents in natural language, making investigations slow and incomplete.
- Existing solutions lack real-time summarization, intelligent event detection, and context-aware insights from live video streams.
- Our project addresses these gaps by developing an AI-powered system that enables real-time detection, summarization, semantic storage, and natural language querying of surveillance data.

Scope

- Real-time monitoring of retail staff and customers using advanced computer vision techniques.
- Event detection and summarization to automatically generate concise, context-aware descriptions.
- Semantic storage of events in a vector database with metadata for long-term retrieval.
- Natural language querying of surveillance data through an AI-powered assistant.
- Support for incident investigation by providing evidence-backed answers instantly.
- Improvement of retail operations through proactive monitoring, loss prevention, and efficiency gains.
- Scalability of the system to be applied in other domains such as smart cities, transportation, and public safety.

Technological Stack

1) Overall Architecture (Pipeline)

- Ingest live CCTV stream → detect & track → trigger event → cut clip → VLM summary → embed & store → RAG Q&A.
- This converts raw video into searchable, time-stamped evidence and natural-language answers for managers.

2) Object Detection — YOLOv8

- A real-time, anchor-free object detector, optimizes classification and box regression separately; uses NMS to remove duplicates.
- High FPS and strong accuracy for people/object classes in retail scenes.
- **Our customizations (value-add):**
  - Domain fine-tuning on retail footage (entrances, cash counters, aisles) for better precision/recall.
  - Dynamic thresholds (class-specific conf/NMS) to reduce false alarms in crowded shelves.
  - Zone masks/ROIs to ignore irrelevant areas (e.g., ceiling reflections).

3) Multi-Object Tracking — Deep SORT

- SORT + deep appearance embeddings; Kalman filter predicts motion; Hungarian algorithm associates detections to tracks; appearance vectors fix ID switches.
- Maintains stable IDs across occlusions and crowd movement.
- **Our customizations (value-add):**
  - Re-ID model tuned on store attire/lighting; longer max_age for short occlusions.
  - Track smoothing and ID persistence rules to keep consistent counts for summaries.
  - Hard negative mining for look-alike uniforms (staff vs shoppers).

4) Event Triggering & Clip Builder

- Rule engine on tracks: zone intrusion, dwell time, backroom entry, after-hours motion, counter crowding, etc.
- Converts continuous video into meaningful incidents.
- **Our customizations (value-add):**
  - Composable rules (AND/OR with time windows) and per-zone sensitivity.
  - Pre/post context windowing (e.g., 4s before + 6s after) to capture the full incident.
  - Multi-track fusion to represent group events (e.g., “two shoppers” rather than separate tracks).

5) Vision-Language Summarization — LLaVA

- A Vision-Language Model that aligns image features with an LLM to produce concise, context-aware text from frames/clips.
- Generates human-readable summaries of what occurred.
- **Our customizations (value-add):**
  - Structured prompt template (Who, What, Where, When, Count) to reduce vagueness.
  - Frame selection: keyframes from peak motion + first/last frames for context.
  - Hallucination guardrails: cross-check counts/objects against detector metadata; if mismatch, auto-revise.
  - Style constraints: objective tone, 1–2 sentences, no identity claims, include timestamp & zone.

6) Text Embeddings — Sentence Transformers

- Converts summaries to dense vectors; supports cosine similarity for semantic search.
- Makes incident memory searchable by meaning, not exact words.
- **Our customizations (value-add):**
  - Domain-adapted embedding choice (balanced speed/quality) + mean-pooling with normalization.
  - Hybrid retrieval: vector similarity + metadata filters (date/zone/event type).
  - Dedup logic to collapse near-identical summaries across adjacent clips.

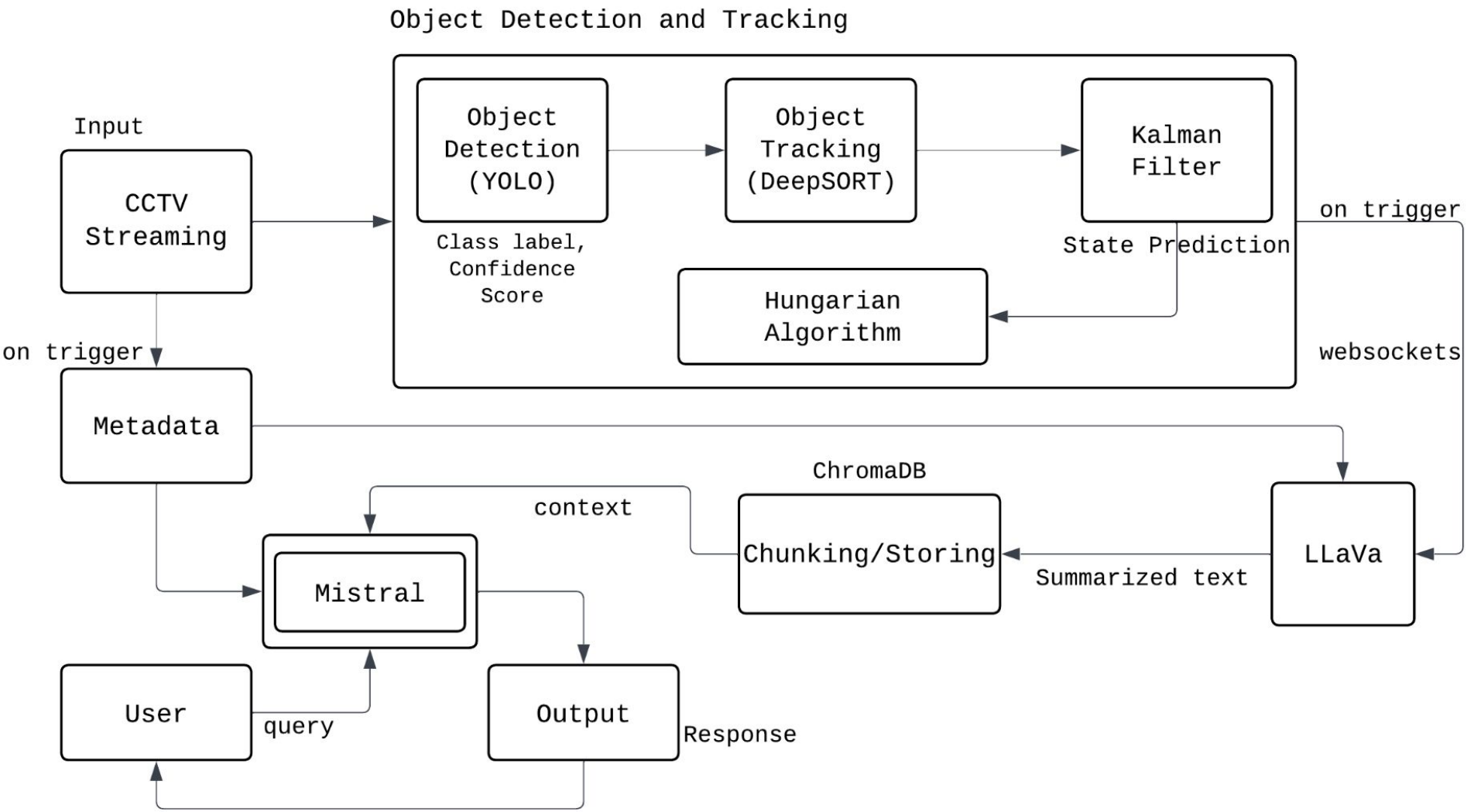
7) Vector Store — ChromaDB

- A vector database storing embeddings + metadata (timestamps, camera ID, zone, track counts).
- Fast similarity search and filtered retrieval for investigations.
- **Our customizations (value-add):**
  - Namespaces per store/camera for quick scoping.
  - Time-aware ranking: boost recent incidents for “what happened just now?” queries.
  - Retention policy with privacy fields (e.g., optional face blur flags).

8) Conversational Q&A — LangChain + Mistral (RAG)

- Retrieval-Augmented Generation: generate multiple query variants → retrieve top-k summaries → LLM composes an evidence-backed answer.
- Lets managers ask natural-language questions and get traceable responses.
- **Our customizations (value-add):**
  - Multi-query expansion and re-ranking for better recall in messy phrasing.
  - Citations: attach event IDs, timestamps, and clip links in the answer.
  - Guardrails: decline identity attribution; stick to observed facts; show confidence score.

Proposed System Architecture



Implementation Status

Frontend Development

- Live camera feed integrated into the user interface.
- “Analyse” button implemented to capture quick snapshots from the live video.

Data Transmission

- Snapshots are converted into byte format and sent to the backend using WebSockets for real-time transfer.

Backend Progress

- Object Detection module (YOLOv8 + Deep SORT) not yet integrated.
- Work has begun on LLaVA (Vision-Language Model), which currently provides basic image descriptions when an image is inputted.

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Thank You...!!